MIL-H-36746A

13 June 1973

SUPERSEDING
MIL-H-36746
16 June 1972

MILITARY SPECIFICATION

HUMIDIFIER, OXYGEN, DIFFUSER TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bubbler-aerosol, dual-action humidifier used in the administration of oxygen therapy.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6515

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STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)
24262 HUMIDIFIER, OXYGEN, Diffuser Type

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

COMPRESSED GAS ASSOCIATION, INC. (CGA)
CGA Pamphlet V-5 DIAMETER -INDEX SAFETY SYSTEM: NON-INTERCHANGEABLE LOW PRESSURE CONNECTIONS FOR MEDICAL GASES, AIR AND SUCTION.

(Application for copies should be addressed to the Compressed Gas Association, Inc., 500 Fifty Avenue, New York, N.Y. 10036.)

* 3.1 Material. The jar, female screw band, and tubing adapter of the humidifier assembly shall be fabricated from clear polycarbonate plastic. The humidifier head shall be fabricated from polycarbonate plastic. The low-pressure seal of the gas inlet shall be fabricated from polytetrafluorethylene (Teflon) or polychlorotrifluorethylene compounds (KEL-F). The wing nut and stem (gas inlet) shall be fabricated from nickel plated, or chromium over nickel plated, copper alloy. The filter component of the syphon tube shall be fabricated from polyurethane suitable for the purpose intended. Nylon shall be prohibited in the fabrication of components that are in direct communication with oxygen. All other components of the humidifier assembly shall be fabricated from corrosion-resistant material, suitable in all respects for the purpose intended. The entire unit (with head removed from jar) shall be capable of withstanding repeated steam sterilization without indication of damage, separation, or deformation when tested as specified in 4.4.1. The entire unit shall also be capable of withstanding ethylene oxide sterilization or cold chemical solutions for disinfecting procedures.

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* 3.2 Style, design and dimensions. Shall be a dual-action humidifier suitable for the administration of oxygen therapy in open systems. The unit shall have two settings. In one setting, the unit shall operate as a "bubbler-type" humidifier. In the other setting, the unit shall operate as a jet-type nebulizer, for medium or high levels of unheated humidification that transfers liquid and oxygen (or air) into a fog. The humidifier shall consist essentially of a head, a jar, a screw band to effect closure of the jar to the head, and a separate tubing adapter. Style, design and dimensions of the humidifier assembly shall be in accordance with DPSC Drawing 24262, revised 9 November 1972.

* 3.2.1 Humidifier head. The humidifier head shall have an outlet port to accommodate NSN 6515-00-055-2155 Tube, Oxygen, Plastic, Large Bore, in a leakproof manner. A wing nut shall be integral to the head and shall be suitable for use with Standard Low-Pressure Connection No. 1240 of CGA Pamphlet V-5. The head shall be easily and directly assembled to oxygen delivery devices such as NSN 6680-00-935-4242 Regulator, Pressure, Medical Gas Administration Apparatus, Lightweight, Oxygen with Flowmeter, without the use of any adapters. An easy operable selector switch, integral to the head, shall be provided to attain bubbler-type humidification, or functions as a jet-type nebulizer. The "bubble" and "jet-type" positions shall be clearly and permanently marked on the external surface of the head. The head shall be gasketed to form a leakproof union with the jar, when assembled by means of the female screw band. A syphon tube shall be securely and permanently attached to the underside of the head. The syphon tube shall be constructed to prevent damage due to possible vibrational effects when the unit is operating. A bubbler-filter shall be attached to the distal end of the water syphon tube and shall be capable of easy removal for replacement or cleaning. The head shall contain a jet-orifice designed to act as jet-type nebulizer with large bore oxygen tubing specified above. The jet-orifice shall be of the self-cleaning type, by switching the unit to the "bubble" position momentarily without interupting service. A pressure relief valve, integral to the head, shall be provided to prevent internal pressure in the humidifier from exceeding 2.5 pounds per square inch gage (psig). A suitable warning device shall be designed in conjunction with the relief valve to provide audible warning should the tubing circuitry become obstructed when the oxygen flow is as low as 3 liters per minute (LPM) when tested as specified in 4.4.2. The audible warning device shall not sound when a normal, unobstructed flow is entering the delivery tubing when tested as specified in 4.4.2.

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* 3.2.2 Jar. The jar shall permit visual observation of liquid level. The jar shall have a liquid capacity of 550 cubic-centimeters (cc) to 600 cc. "REFILL" and "FILL TO THIS LINE" permanent markings shall appear on the jar. The jar shall be threaded to effect easy assembly and removal to the humidifier head and mate with the female screw band. The jar shall be capable of withstanding a minimum internal pressure of 200 psig without bursting when tested as specified in 4.4.3.

3.2.3 Separate tubing adapter. Each humidifier shall be furnished with a separate tapered adapter designed for use with catheters having inside diameter ranging from 3/16 inch to 5/16 inch. The adapter shall be furnished with a suitable "O" ring that shall insure leakproof performance. In addition, a suitable retainer shall be attached to the adapter that can be readily secured to the outlet port of the head.

3.3 Flow rates. When tested as specified in 4.4.4, the humidifier head (jar removed) shall allow a minimum flow rate of 10 LPM in the "bubble" position, and a minimum flow rate of 12 LPM in the "jet" position. Humidifiers that fail this test shall be rejected.

* 3.4 Humidifier performance. The humidifier shall be capable of maintaining the following relative humidities at body temperature (98.6 degrees Fahrenheit):

Position	Relative Humidity
As a Bubble-Type Humidifier:	43 percent minimum at 3 LPM flow to 28 percent minimum at 10 LPM flow.
As a Jet-Type Humidifier:	40 percent minimum at 3 LPM flow to 67 percent minimum at 10 LPM flow.

The relative humidities shall be calculated by determining the weight of water expended (with oxygen admitted to the humidifier at 70° F and 29 inches of mercury) and the resultant molal humidity of issuant gas. The vapor pressure of the issuant gas divided by its partial pressure of saturated vapor at 98.6° F, shall be defined as the relative humidity.

3.5 Functionability of humidifier. The unit shall be suitable for use with nonpressure compensated oxygen flowmeters, such as specified in 3.2.1, without causing hazardous effects with respect to intended safe oxygen, or compressed air, flow rates.

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3.6 Service Data. The contractor shall furnish, with each humidifier, one copy each of one or more booklets which contain complete instruction for installation, operation, and maintenance and lists of component parts. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved service data to the Contracting Officer for distribution. Should the supplier have furnished acceptable service data for the identical item within two years preceding the date of contract and propose to furnish service data identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of the drafts and 15 additional copies shall be at the option of the government.

3.7 Identification marking. The humidifier shall be clearly and permanently marked with the contractor's name or registered trademark.

3.8 Workmanship. Workmanship shall be first class throughout. The humidifiers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the copper alloys and plastics (head components) may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table I. The unit of product for examination purposes shall be one humidifier.

Table I.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one humidifier.

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Table II. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Humidifier	Sterilization test	3.1	4.4.1	*	***
	Dimensions	3.2	---	S-1	2.5
	Relief valve performance	3.2.1	4.4.2	S-1	1.5
	Bursting pressure test for jar	3.2.2	4.4.3	**	***
	Flow rate-test	3.3	4.4.4	S-1	1.5

* Two complete humidifiers shall be tested for each contracted quantity.
** Two jars shall be tested for each contracted quantity.
*** No defects permitted.

* 4.3 Examination. The humidifiers shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any humidifiers in the sample containing one or more defects shall be rejected and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Humidifier not of the dual-action (bubble and jet) type.
102	Absence of relief valve.
103	Head not readily assembled or removed from jar.
104	Wing nut (low-pressure gas connection) not integral to head.
105	Selector switch not capable of easy operation.
106	Bubble and jet positions not permanently identified on head.
107	Jar does not possess markings as specified in 3.2.2.
108	Wing nut not capable of easy assembly to, or removal from, standard oxygen connection specified in 3.2.1.
109	Syphon tube assembly does not contain flexible connection.
110	Syphon tube does not possess replaceable filter at distal end.
111	Syphon tube assembly not permanently secured to underside of head.
112	Syphon tube filter in contact with bottom of jar.
113	Separate tapered tubing adapter, with "O" ring and retainer, not furnished.
114	Components not free from cracks, pits, holes, burrs, slivers, sharp edges, scale, corrosion, and excessive flash.
115	Service data not furnished.

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Table III. Classification of Defects

Categories	Defects
Minor	
201	Humidifiers not marked with contractor's name or registered trademark.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Sterilization test. All humidifer components, with head disassembled from the jar, shall be placed in a suitable sterilized (container) at a temperature of 250° F for a period of at least 10 minutes and not to exceed 20 minutes. The humidifier components shall then be removed from the sterilizer and allowed to cool to room temperature. This test shall be performed a total of 10 times on the same sample unit.

* 4.4.2 Relief valve performance test. A pneumatic circuit consisting essentially of an adjustable pressure regulator, pressure gage (with at least 100 psig capacity), precision flowmeter (with at least 15 LPM capacity), restriction device, mercury monometer, and the humidifier assembly (jar filled with water at "FILL TO THIS LINE" marking) complete with small tubing adapter inserted into the outlet port of the humidifier, shall be used to test performance of the relief valve. Suitable tubing (minimum inside diameter 3/16 inch) shall be connected to the tubing adapter, and the mercury manometer, in a leakproof manner. Compressed air, or oxygen, shall be internally admitted to the humidifier inlet. The regulator shall be adjusted to allow a flow of 3 LPM to enter the humidifier (with the humidifier set in the jet position) and the outlet restricted until a differential pressure from 4.0 to 5.0 inches of mercury (in Hg) is created in the humidifier. The relief valve shall be observed for satisfactory performance as evidenced by an audible warning sound; the relief valve shall continue to oscillate.

Inlet flow shall then be adjusted to 10 LPM and the relief valve shall be tested as above. The relief valve shall then be blocked by placing a finger over the opening on the cap; the unit shall maintain a steady differential pressure of at least 12 in Hg (at 3 LPM). Should unit not maintain at least 12 in Hg (at 3 LPM), it shall be rejected. Absence of audible warming emitted by the relief valve with an unobstructed flow of 3 LPM shall indicate passage of the unobstructed flow test.

The circuitry shown in Figure 1, is indicated only as a guide to perform the relief valve performance test.

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* 4.4.3 Bursting pressure test for jar. The head of the humidifier shall be removed from the jar. The jar opening shall be fitted with a secured leakproof-device that shall have a liquid inlet member. Water, or other suitable liquid, at a minimum internal pressure of 200 psig shall be admitted to the jar and the pressure maintained for at least 3 minutes. Care shall be taken in the performance of this test to shield the operator from possible bursting material.

* 4.4.4 Flow rate test. A pneumatic circuit consisting of an adjustable pressure regulator, pressure gage (100 psig minimum capacity), precision flowmeter, humidifier head with jar and filter assembly removed shall be used to test flow rates. Unit shall be set to the bubble position, and air admitted to the inlet of the humidifier head at a minimum pressure of 50 psig and the flowmeter adjusted to obtain maximum flow. Units that do not indicate a minimum flow of 10 LPM shall be rejected. The humidifier head shall then be set to the jet position and the flow meter again adjusted to obtain maximum flow. Units that do not indicate a minimum flow of 12 LPM shall be rejected. Flexible tubing of 1/8 inch inside diameter shall be securely attached to the distal end of the syphon tube and to a U-tube water manometer in a leakproof manner. The selector switch shall be set to the "jet" position and the flow adjusted to 2.5 LPM. Units that do not indicate a minimum differential pressure of 6 inches of water shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One humidifier, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be clean, wrapped in neutral tissue or other suitable anticorrosive material. The unit shall be further packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable or PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements as specified in the applicable table in PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with paragraph 30.4 of PPP-B-636.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3, PPP-B-601, domestic type, PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636. Special requirements as specified in the applicable table in PPP-B-636 shall apply for fiberboard boxes.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The dual-action humidifiers are intended for the administration of oxygen therapy in which the applicator generates the liquid and oxygen or air into a fog.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following item appear- in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-469-6573	HUMIDIFIER, OXYGEN, Diffuser Type

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1716

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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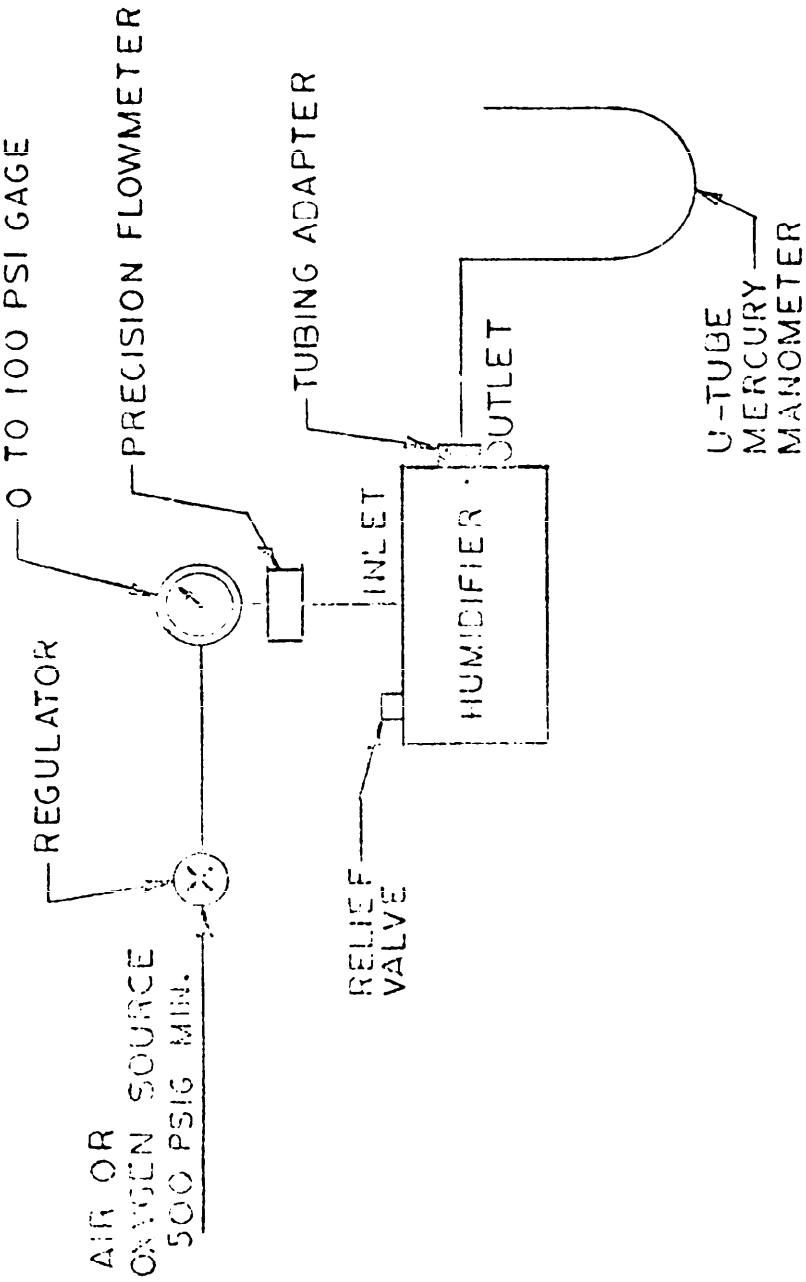


FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

MIL-H-36746A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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